Component-and-connect patterns:

- Broker
- Model-view Controller
- Pipe and Filter
- Client-Server
- **Peer-to-Peer**
- Service-Oriented-Architecture
- Publish-Subscribe
- Shared-Data

Peer-to-Peer Pattern

- **Context:** Distributed computational entities—each of which is considered equally important in terms of initiating an interaction and each of which provides its own resources—need to cooperate and collaborate to provide a service to a distributed community of users.
- **Problem:** How can a set of “equal” distributed computational entities be connected to each other via a common protocol so that they can organize and share their services with high availability and scalability?
- **Solution:** In the peer-to-peer (P2P) pattern, components directly interact as peers. All peers are “equal” and no peer or group of peers can be critical for the health of the system. Peer-to-peer communication is typically a request/reply interaction without the asymmetry found in the client-server pattern.
  - That is, any component can, in principle, interact with any other component by requesting its services.
  - The interaction may be initiated by either party—that is, in client-server terms, each peer component is both a client and a server.

Peer-to-Peer Example

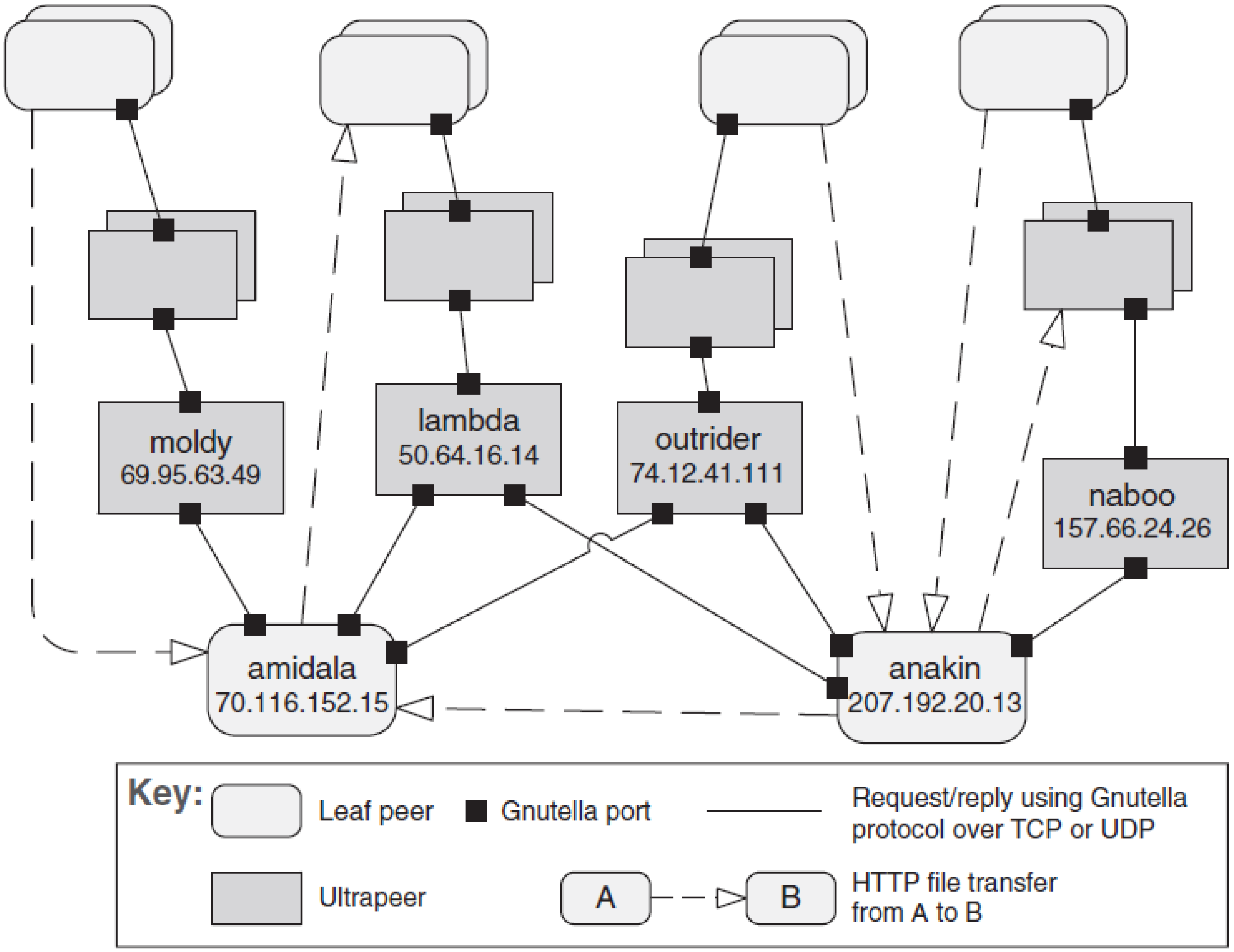

FIGURE 13.10 A peer-to-peer view of a Gnutella network using an informal C&C notation. For brevity, only a few peers are identified. Each of the identified leaf peers uploads and downloads files directly from other peers.

Peer-to-Peer Solution - 1

- Overview: Computation is achieved by cooperating peers that request service from and provide services to one another across a network.

- Elements:
  - *Peer*, which is an independent component running on a network node. Special peer components can provide routing, indexing, and peer search capability.
  - *Request/reply connector*, which is used to connect to the peer network, search for other peers, and invoke services from other peers.

- Relations: The relation associates peers with their connectors. Attachments may change at runtime.

Peer-to-Peer Solution - 2

- Peers first connect to the peer-to-peer network on which they discover other peers they can interact with, and then initiate actions to achieve their computation by cooperating with other peers by requesting services.

- Often a peer's search for another peer is propagated from one peer to its connected peers for a limited number of hops.

- A peer-to-peer architecture may have specialized peer nodes (called supernodes) that have indexing or routing capabilities and allow a regular peer's search to reach a larger number of peers.

Peer-to-Peer Solution - 3

- Peers can be added and removed from the peer-to-peer network with no significant impact, resulting in great scalability for the whole system. This provides **flexibility** for deploying the system across a highly distributed platform.

- Typically multiple peers have overlapping capabilities, such as providing access to the same data or providing equivalent services.
  - Thus, a peer acting as client can collaborate with multiple peers acting as servers to complete a certain task.
  - If one of these multiple peers becomes unavailable, the others can still provide the services to complete the task. The result is **improved overall availability**.

Peer-to-Peer Solution - 4

- **Performance** advantages:
  - The load on any given peer component acting as a server is reduced,
  - the responsibilities that might have required more server capacity and infrastructure to support it are distributed.
  - This can decrease the need for other communication for updating data and for central server storage, but at the expense of storing the data locally.

Peer-to-Peer Solution - 5

- **Constraints**: Restrictions may be placed on the following:
  - The number of allowable attachments to any given peer
  - The number of hops used for searching for a peer
  - Which peers know about which other peers
  - Some P2P networks are organized with star topologies, in which peers only connect to supernodes.

- **Weaknesses**:
  - Managing security, data consistency, data/service availability, backup, and recovery are all more complex.
    - Because peer-to-peer systems are decentralized
  - Small peer-to-peer systems may not be able to consistently achieve quality goals such as performance and availability.

Usages of Peer-to-Peer

- Peer-to-peer computing is often used in distributed computing applications such as:
  - File sharing,
  - Instant messaging,
- Examples of peer-to-peer systems include file-sharing networks such as BitTorrent, and instant messaging and VoIP applications such as Skype.